

① Tabbellor ordnades e uttrycks med Karnaugh

1.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

C \ AB	AB			
	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$F = C + AB$$

2.

A	B	C	G
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

C \ AB	AB			
	00	01	11	10
0	0	1	0	0
1	1	1	1	1

$$G = C + \bar{A}B$$

3.

A	B	C	D	H
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

C \ AB	AB			
	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$H = \bar{B} + AC$$

4.

A	B	C	I
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

C \ AB	AB			
	00	01	11	10
0	0	1	0	1
1	0	1	0	1

$$I = A \oplus B$$

A	B	C	D	J
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

CD \ AB	00	01	11	10
00	0	0	0	1
01	0	0	0	1
11	1	1	1	1
10	0	0	0	1

$$J = A\bar{B} + CD$$

5.

② Simplificação com álgebra booleana

$$6. F = AB + \cancel{A\bar{B}} + \cancel{\bar{A}B}$$

$$F = AB$$

$$7. F = (A+B)(A+B)$$

$$(A+A)(A+B)(B+A)(B+B)$$

$$F = A$$

$$8. F = \cancel{A\bar{B}C} + \cancel{A\bar{B}C} + \cancel{AC\bar{B}} + \cancel{AC\bar{B}}$$

$$F = AC$$

$$9. F = \cancel{\bar{A}B} + \cancel{\bar{A}B} + \bar{A}\bar{B}C$$

$$F = \bar{A} + \bar{B}C$$

$$10. F = A + \bar{A}B$$

$$(A+A)(A+B)$$

$$F = A + B$$

11. $F = (A+B+C)(A+B+C)$

$(A+A)(A+\bar{B})(A+C)(\bar{B}+A)(\bar{B}+\bar{B})(\bar{B}+C)(C+A)(C+\bar{B})(C+C)$

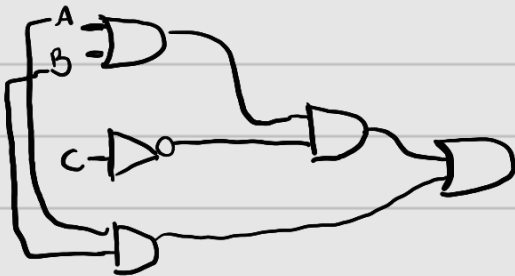
$F = A+C$

12. $F = AB + \bar{A}C + BC$

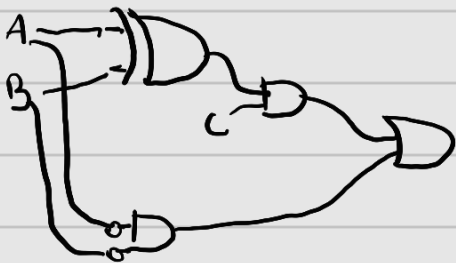
$F = AB + \bar{A}C$

③ Expressões para circuitos

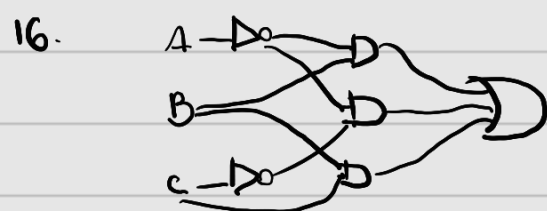
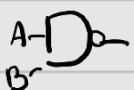
13. $F = (A+B)\bar{C} + AB$



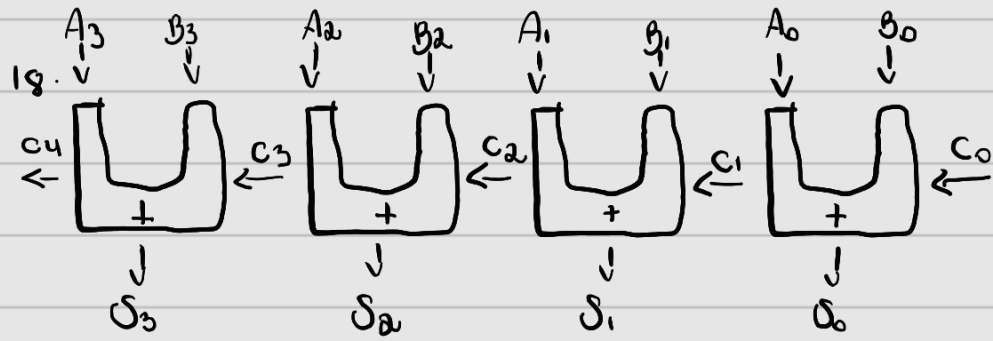
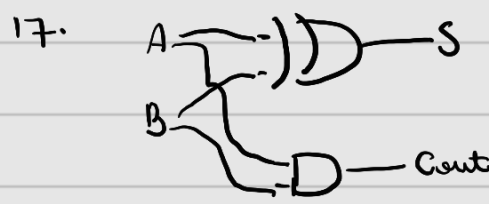
14. $F = (A \oplus B)C + \bar{A}\bar{B}$



15. $F = \overline{AB}$

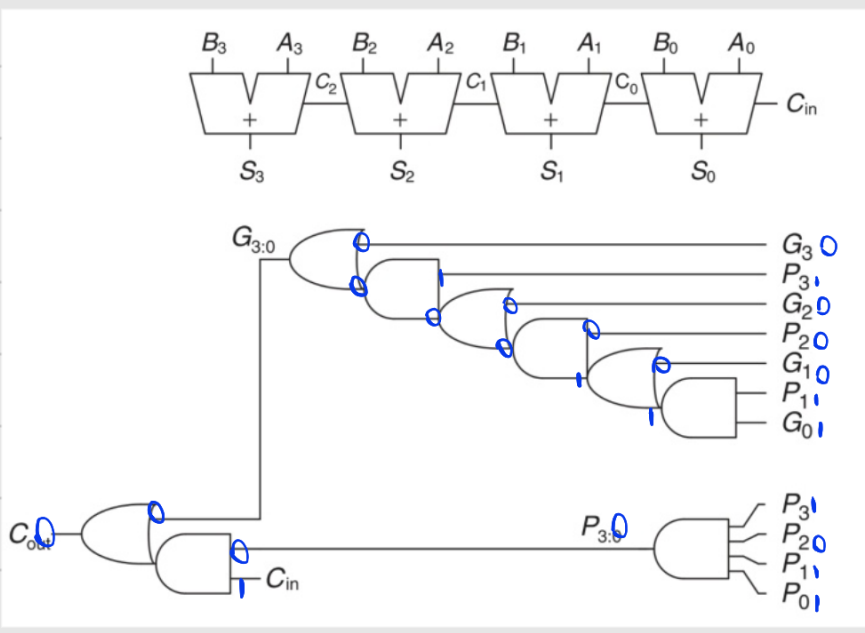


4. Semantics



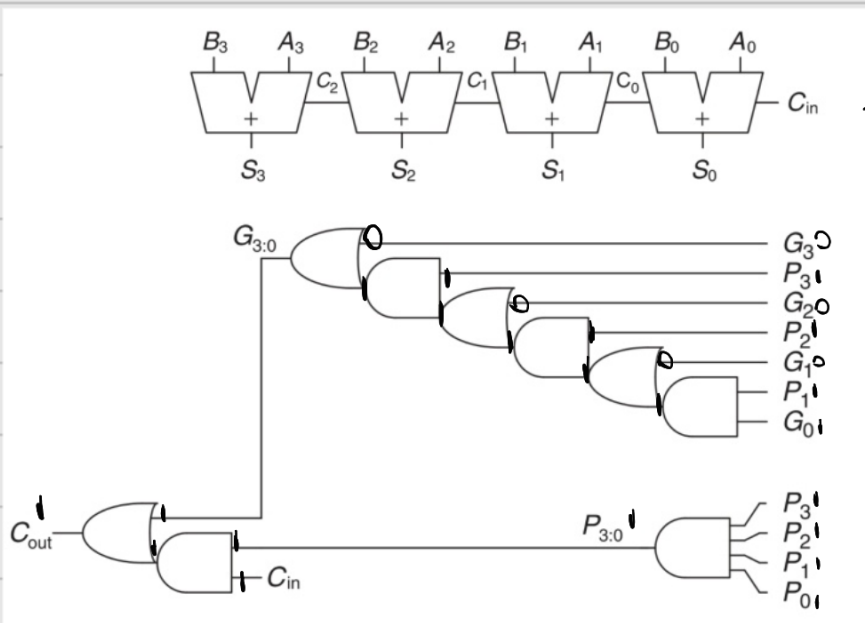
19.

(a)



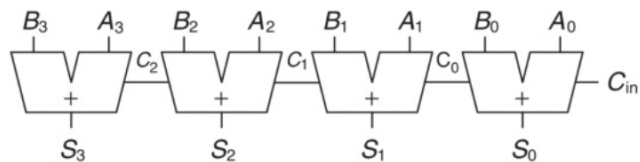
3 2 1 0
1 0 1 1
+ 0 0 0 1 cin = 1

(b)

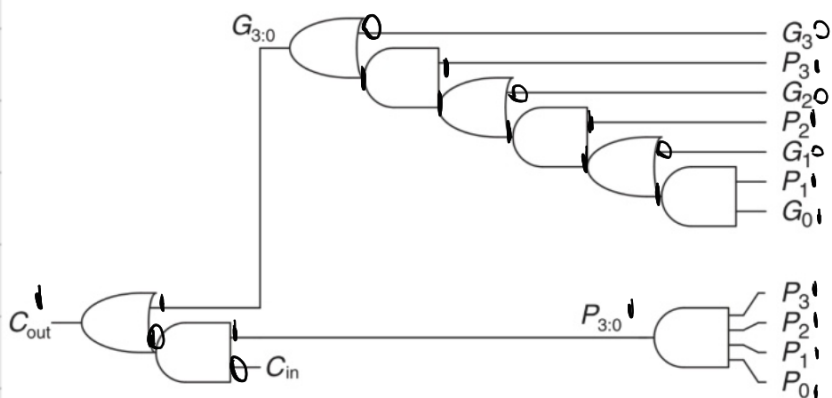


3 2 1 0
1 0 1 1
+ 0 1 0 1 cin = 1

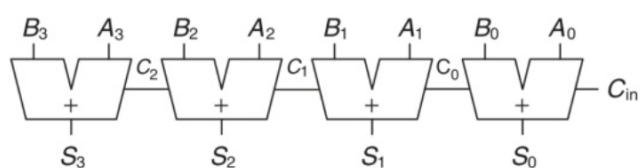
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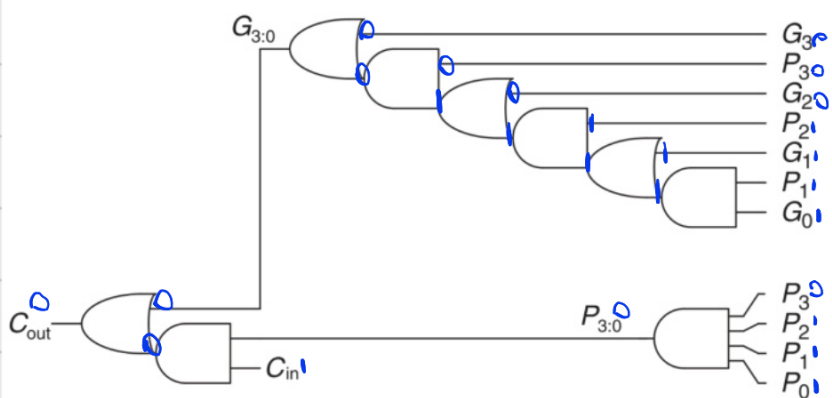
$$\begin{array}{r} 3210 \\ 1011 \\ + 0101 \\ \hline \text{cin} = 0 \end{array}$$



©



$$\begin{array}{r} 3210 \\ 0011 \\ + 0111 \\ \hline \text{cin} = 1 \end{array}$$



⑤ Circuitos para expressões

20. $Y = A\bar{B} + AB + AC$
 $Y = A$

21. $G = A\bar{C} + B\bar{C} + AC$
 $A + BC$

22. $X = A \oplus B$

23.

